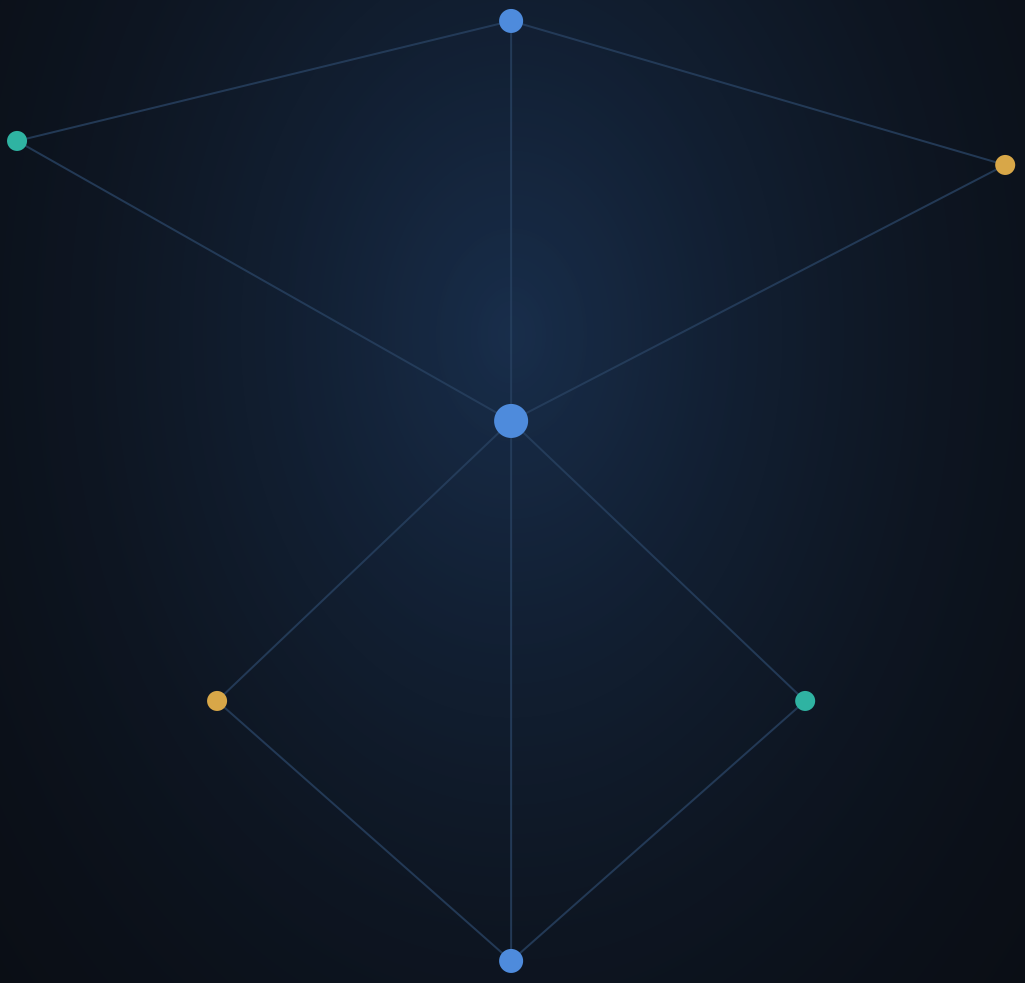
Data Intelligence OS



Product Requirements Document · MVP

Data Intelligence OS

AI-Powered Trusted Dataset Platform

Help organisations discover, understand, validate and create trusted datasets faster using AI-assisted data engineering. This document defines the MVP only.

VERSION	DATE	OWNER	AUDIENCE	STATUS
1.0	October 2026	Product · PyrunAI	Exec · Product · Engineering	For review

SECTION 01

Executive Summary

Data Intelligence OS is an AI-assisted platform that turns a database connection into a documented, validated dataset in hours rather than weeks — and proves the result is correct before anyone uses it.

What it is

A layer that sits on top of systems an organisation already runs. It connects read-only, discovers what is there, measures quality, finds and verifies relationships, assembles datasets with the arithmetic checked, and generates the documentation and lineage automatically.

What it is not

Not an AutoML platform — model training is out of scope. Not a data catalogue — it builds and verifies rather than describes. Not a traditional ETL tool — it does not compete on moving data between systems.

The problem

A business request for a dataset takes four to six weeks. Most of that time is not engineering — it is discovery, guesswork and rework: finding the right tables, inferring how they join, cleaning values, and discovering late that a join multiplied the amounts. Reconciliation is manual and usually skipped. Documentation rarely happens. When the engineer leaves, the reasoning leaves too.

PRIMARY GOAL **Hours** dataset creation, down from days or weeks

TARGET USERS **4** engineer, analyst, scientist, architect

MVP MODULES **9** scope-locked; everything else deferred

CONNECTORS **4** SQL Server, PostgreSQL, CSV, Excel

Target users

- **Data Engineer** — primary user, builds the datasets
- **Business Analyst** — consumes certified datasets
- **Data Scientist** — needs reproducible, documented inputs
- **Data Architect** — approves standards and the landscape view

Business value

- Weeks of discovery and verification removed from every request
- Quality and completeness become reportable numbers, not assertions
- Documentation generated, so it cannot drift from the pipeline
- The reasoning behind each dataset is retained when staff change

Scope discipline. This PRD covers the MVP only. Nine modules are in; eight capabilities are explicitly deferred to later releases and listed in Section 05. Adding anything to this scope requires a written change to this document.

SECTION 02

Problem Statement

Six problems define the MVP. Each one maps to a module, so nothing is built without a problem behind it.

PROBLEM	WHAT IT LOOKS LIKE TODAY	HOW DATA INTELLIGENCE OS SOLVES IT
Data scattered across systems	A customer view needs core banking, a CRM, a payments warehouse and three spreadsheets. No one person has access to all of them.	One read-only connection layer across SQL Server, PostgreSQL, CSV and Excel, with everything surfaced in a single landscape view.
Poor documentation	Fewer than 15% of columns carry a usable description. Tables are named `T_CUST_M_01`.	Documentation generated from the catalogue and the build record, so it exists by default and cannot drift.
Unknown relationships	Foreign keys were dropped years ago. Two columns named `customer_id` often hold different identifiers.	Candidate joins proposed by AI, then measured on live data. Match rate and fan-out reported before anything is built.
Slow dataset creation	Four to six weeks per request, most of it discovery and rework rather than engineering.	Discovery and profiling in minutes; verified join paths; a visual builder with live preview at every step.
Dependency on data engineers	Every business question becomes a ticket. The engineering queue limits how fast the organisation can decide.	Analysts browse and reuse datasets that are already built, documented and validated, without raising a ticket.
Data trust issues	Three reports show three revenue figures. A join silently multiplies a total and nobody notices for weeks.	Validation on completeness, consistency and quality before publication, with the result recorded against the dataset.

The failure mode that justifies the product. Joining a loan table to its repayment history multiplies each loan by its number of EMI rows. Summing the loan amount after that join returns a figure many times too large. Nothing errors — the query succeeds and the report renders. The MVP treats this class of error as a design-time check rather than a production surprise.

SECTION 03

Target Users

Four personas. The Data Engineer is the primary user and the one the MVP is optimised for; the other three are served through what the engineer produces.

Primary

Data Engineer

GOALS

Deliver correct datasets quickly. Keep pipelines reliable. Spend time on engineering rather than on discovery.

PAIN POINTS

Weeks lost finding tables and verifying joins. Reconciliation skipped under deadline pressure. Documentation never written. Sole holder of undocumented knowledge.

BENEFITS

Profiling in minutes. Join candidates measured, not guessed. Fan-out caught at design time. Documentation produced automatically.

Consumer

Business Analyst

GOALS

Answer business questions without waiting weeks. Defend the numbers presented to stakeholders.

PAIN POINTS

Cannot tell an authoritative table from an abandoned copy. No way to know whether a dataset was validated. Every request becomes a ticket.

BENEFITS

Browse datasets that are already documented and validated, with the data dictionary and quality score attached.

Consumer

Data Scientist

GOALS

Get to modelling faster. Reproduce a dataset exactly as it was six months ago.

PAIN POINTS

40–60% of project time lost to data preparation. No record of how a training set was assembled. Quality problems discovered after training.

BENEFITS

Documented datasets with lineage and a quality score, so preparation starts from a known-good base rather than from raw tables.

Approver

Data Architect

GOALS

Keep the real landscape understood and standards applied in practice, not just on paper.

PAIN POINTS

The documented architecture diverged from reality years ago. No reliable map of what connects to what.

BENEFITS

A live landscape view derived from the systems themselves, with relationships measured rather than declared.

SECTION 04

MVP Scope — modules 1 to 5

Nine modules are in scope. Nothing else ships in v1. Each entry states what is included and what is explicitly excluded, so the boundary is unambiguous during build.

#	MODULE	IN SCOPE	EXPLICITLY EXCLUDED FROM MVP	PRIORITY
01	**Data Source Connectivity**	Read-only connections to SQL Server and PostgreSQL. CSV and Excel upload. Credentials in a vault. Connection test and schema filter.	MySQL, Oracle, Snowflake, REST APIs, SAP, streaming sources.	P0
02	**Data Landscape Explorer**	Tables, columns, data types, row counts. Relationship graph with measured match rates. Click through to table detail.	Cross-source relationships, business domains, usage popularity.	P0
03	**AI Discovery Center**	Schema discovery, relationship suggestions with confidence scores, AI-drafted table and column descriptions for human approval.	Autonomous acceptance of suggestions. Learning from past decisions.	P0
04	**Data Health Center**	Null detection, duplicate detection, a composite quality score per table and per column, with affected row counts.	Freshness checks, outlier rules beyond a default, scheduled re-profiling, alerting.	P0
05	**Visual Dataset Builder**	Multi-table joins on verified paths, filters, declared grain with fan-out warning, live dataset preview with row and column counts.	Drag-and-drop canvas, cross-source joins, nested sub-queries.	P0

Why these five first

They form the shortest path from a connection to a dataset a user can see. Without all five, there is no demonstrable product; with them, every later module adds trust rather than function.

The non-negotiable behaviour

Module 05 must warn before a join multiplies rows. This single check is the clearest differentiator in the MVP and must not be deferred for schedule reasons.

Human in the loop

Module 03 proposes; it never applies. Every AI suggestion carries a confidence score and requires an explicit human action before it affects a dataset.

SECTION 04 · CONTINUED

MVP Scope — modules 6 to 9

#	MODULE	IN SCOPE	EXPLICITLY EXCLUDED FROM MVP	PRIORITY
06	**Transformation Studio**	Remove nulls, remove duplicates, datatype conversion, derived columns. Rows before and after shown for every step. Steps saved with the dataset and re-runnable.	Merge and split columns, text standardisation, custom code, Python or SQL snippets.	P0
07	**Validation Center**	Completeness score, consistency score and an overall quality validation with pass, warning or fail per check.	Reconciliation against source totals, freshness, integrity scoring, user-defined rule builder.	P0
08	**Documentation Center**	Data dictionary per dataset and auto-generated dataset documentation, exportable. Regenerated on every build.	Business glossary, owner approval workflow, publishing to an external catalogue.	P1
09	**Basic Lineage**	Source → dataset → transformations, as a readable flow with each step clickable.	Column-level lineage, lineage to reports or models, lineage through stored procedures.	P1

SECTION 05

Out of Scope for MVP

Deferred to later releases. Listed here so the boundary is explicit and so the team can answer scope questions without a meeting.

Governance

Audit logs, access control, policies, compliance reporting. Needs authentication and user roles first.

Dataset Marketplace

Sharing, cloning and certification. Low value until a customer has many datasets.

Knowledge Layer

Versioned business definitions the AI reads before the schema. High value, Phase 2.

Data Contracts

Expected versus delivered schema, drift detection and impact. Phase 2.

Root Cause Analysis

Requires monitoring history to analyse. Cannot be built meaningfully before it.

Auto-Healing

Automatic remediation. Out of scope on principle until trust is established.

Advanced Monitoring

Scheduler, metrics store and alert routing — three systems, not one feature.

Enterprise Approvals

Multi-step sign-off with second approver. Needs roles and audit first.

One dependency worth stating plainly. Four of these eight deferred items are blocked by the same prerequisite: authentication, users and roles. That work is small, unglamorous and should be scheduled in the MVP build even though it is not a user-facing module — otherwise Phase 2 cannot start.

SECTION 06

End-to-End User Workflow

Nine steps from connecting a source to a trusted dataset. Steps 3 and 7 are the review points where a human must act before the flow continues.

1. **Connect source**
   Read-only connection to SQL Server or PostgreSQL, or a CSV / Excel upload.
2. **Discover tables**
   Tables, columns, types and row counts inventoried automatically.
3. **Review relationships** — Review point
   AI proposes joins with confidence; the platform measures match rate on live data. User confirms.
4. **Check data health**
   Nulls, duplicates and a quality score per table and column, with affected row counts.
5. **Build dataset**
   Multi-table join on confirmed paths, filters applied, live preview with row and column counts.
6. **Apply transformations**
   Remove nulls, remove duplicates, convert datatypes, add derived columns. Before and after shown.
7. **Validate dataset** — Review point
   Completeness, consistency and overall quality scored. Pass, warning or fail per check.
8. **Generate documentation**
   Data dictionary and dataset documentation produced from the build record.
9. **Trusted dataset ready**
   Published with its quality score, validation result, documentation and lineage attached.

Steps 1–4 · Understand

Connect, discover, review relationships and check health. Nothing is written and no dataset exists yet. Output is knowledge the user can act on.

Steps 5–6 · Build

Assemble the dataset and apply transformations. Every operation shows rows before and after, and every step is saved so the build can be repeated.

Steps 7–9 · Prove

Validate, document and publish. A dataset that fails validation can still be published in the MVP, but it is marked and the failure is recorded against it.

SECTION 07

Key User Stories

Two to three stories per module, written to be testable. Modules 1 to 5 on this page; 6 to 9 overleaf.

01 · CONNECTIVITY

As a **Data Engineer**, I want to connect to SQL Server and PostgreSQL with read-only credentials, so that I can explore production data without any risk of writing to it.

01 · CONNECTIVITY

As a **Data Engineer**, I want to upload a CSV or Excel file and have it typed automatically, so that I can combine file-based data with database tables.

02 · LANDSCAPE

As a **Data Architect**, I want to see every table and column in one map, so that I can understand the real landscape rather than the outdated diagram.

02 · LANDSCAPE

As a **Data Engineer**, I want to see the measured match rate on each relationship, so that I know which joins are safe before I design anything.

03 · AI DISCOVERY

As a **Data Engineer**, I want relationship suggestions with a confidence score, so that I can start from twelve candidates instead of inspecting two thousand column pairs.

03 · AI DISCOVERY

As a **Business Analyst**, I want AI-drafted descriptions on tables and columns, so that I can understand what a field means without asking an engineer.

03 · AI DISCOVERY

As a **Data Architect**, I want to approve or reject each AI description, so that the catalogue reflects what the business agrees is true.

04 · DATA HEALTH

As a **Data Engineer**, I want null and duplicate counts per column, so that I can judge whether a table is usable before I build on it.

04 · DATA HEALTH

As a **Data Scientist**, I want a quality score on every table, so that I can rule out unsuitable sources in minutes rather than after a failed model run.

05 · DATASET BUILDER

As a **Data Engineer**, I want to join several tables and see a live preview, so that I can confirm the result is what I expected at each step.

05 · DATASET BUILDER

As a **Data Engineer**, I want a warning when a join will multiply rows, so that I never publish an inflated total by accident.

05 · DATASET BUILDER

As a **Business Analyst**, I want to apply filters without writing SQL, so that I can narrow a dataset to my segment on my own.

SECTION 07 · CONTINUED

Key User Stories — modules 6 to 9

06 · TRANSFORMATION

As a **Data Engineer**, I want to remove nulls and duplicates and see the rows before and after, so that I can confirm the change did what I intended.

06 · TRANSFORMATION

As a **Data Engineer**, I want to convert a column's datatype, so that date and numeric comparisons behave correctly downstream.

06 · TRANSFORMATION

As a **Business Analyst**, I want to add a derived column from a simple formula, so that I can create the measure I need without an engineering ticket.

07 · VALIDATION

As a **Data Engineer**, I want completeness and consistency scored automatically, so that I can prove the dataset's state rather than assert it.

07 · VALIDATION

As a **Data Scientist**, I want to see which validation checks failed and why, so that I can decide whether the dataset is fit for my use case.

08 · DOCUMENTATION

As a **Business Analyst**, I want a data dictionary for every dataset, so that I can interpret a column without guessing.

08 · DOCUMENTATION

As a **Data Engineer**, I want documentation regenerated on every build, so that it never drifts from what the dataset actually contains.

09 · LINEAGE

As a **Data Architect**, I want to trace a dataset back to its source tables and transformations, so that I can answer where a number came from.

SECTION 08

Success Criteria

KPI	BASELINE	MVP TARGET	MEASUREMENT
Dataset creation time	4–6 weeks	Under 2 days	Elapsed time from connection to a published dataset, median across the pilot
User adoption	—	≥ 60% weekly active	Licensed data engineers using the platform at least once a week by month three
Documentation coverage	< 15% of columns	100% of published	Published dataset columns carrying a generated or approved description
Data quality improvement	Not measured	+10 points	Composite quality score on a dataset, before and after transformations
Dataset reuse	Near zero	≥ 30%	New datasets created by copying an existing definition rather than starting fresh

Not a success metric. Headcount reduction. The MVP recovers engineer time and redirects it; it does not remove roles. Claiming otherwise produces a business case that fails at renewal.

SECTION 09

MVP Release Criteria

Every item must be complete before launch. An unticked box is a blocker, not a risk to be accepted.

FUNCTIONAL

- [ ] **All nine MVP modules** are reachable from the navigation and function end to end.
- [ ] **Four connectors work** — SQL Server, PostgreSQL, CSV and Excel — each verified against a real source.
- [ ] **Fan-out warning fires** on a join that multiplies rows, proven with a regression test.
- [ ] **Transformations are saved** with the dataset and reproduce the same output when re-run.
- [ ] **Documentation regenerates** automatically on every build.
- [ ] **Lineage renders** source through transformations to dataset, with each step clickable.

QUALITY

- [ ] **Automated test suite green**, with regression coverage on join correctness and grain.
- [ ] **Tested at realistic scale** — at least one source with 10 M+ rows profiled and built.
- [ ] **No P0 or P1 defects open.**
- [ ] **Error states designed** for failed connections, timeouts and unparseable files.

PLATFORM & SECURITY

- [ ] **Authentication and user roles** implemented. Not a module, but a launch blocker.
- [ ] **Credentials stored in a vault**, never in the database or in logs.
- [ ] **All connections read-only**, verified by attempting a write and confirming it fails.
- [ ] **No row-level data in any AI payload**, proven by an automated test.
- [ ] **Database migrations** managed and reversible.

READINESS

- [ ] **One pilot customer** has built a real dataset end to end and confirmed the output.
- [ ] **Onboarding path** from first login to first dataset in under 30 minutes.
- [ ] **Support and escalation** route defined.
- [ ] **KPI instrumentation live** for all five success metrics in Section 08.

Definition of done for the MVP. A new user connects a database they have never seen, and within one working day produces a dataset that is joined correctly, cleaned, validated, documented and traceable — without writing SQL and without asking anyone for help.